

Program : Diploma in Renewable Energy	
Course Code : 3298	Course Title: Electrical & Renewable Measurements Lab
Semester : 3	Credits: 1.5
Course Category: Programme Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To develop measurement systems for various electrical circuits.
- To develop measurement systems for various renewable systems.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basics of Electrical Measuring Instruments.		Fundamentals of Electrical & Electronics Engineering Lab	2
Concepts of Electrical Power and Energy		Fundamentals of Electrical & Electronics Engineering	

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify standard tests for measurement of resistance.	12	Applying
CO2	Choose various methods for measuring electric power and energy.	9	Applying
CO3	Apply standard calibration procedures to examine accuracy of electrical meters.	9	Applying
CO4	Identify the experimental procedures for the measurement of solar radiation and wind energy.	9	Applying

	Lab Test	6	
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CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Identify standard tests for measurement of resistance.		
M1.01	Sketch the front panel of various electrical measuring instruments- Ammeters (MC&MC) , Voltmeters (MC&MC), Wattmeter and Energy meter	3	Understanding
M1.02	Develop circuits to measure low and medium resistance using the voltmeter-ammeter method.	3	Applying
M1.03	Apply Kelvin's double bridge method for the measurement of low resistance.	3	Applying
M1.04	Apply Wheatstone's bridge method for the measurement of medium resistance.	3	Applying
CO2	Choose various methods for measuring electric power and energy.		
M2.01	Select wattmeter to measure power consumption of a given load.	3	Applying
M2.02	Choose single phase induction type and digital energy meters to measure energy consumption of a given load.	3	Applying

M2.03	Develop a circuit arrangement to measure three phase power and power factor using 2 wattmeter method.	3	Applying
	Lab Test – I	3	
CO3	Apply standard calibration procedures to examine accuracy of electrical meters.		
M3.01	Develop a circuit to calibrate wattmeter at UPF by direct loading.	3	Applying
M3.02	Develop a circuit to calibrate single phase Energy Meter at UPF by direct loading.	3	Applying
M3.03	Develop a circuit to calibrate single phase Energy Meter at UPF by phantom loading.	3	Applying
C04	Identify the experimental procedures for the measurement of solar radiation and wind energy.		
M4.01	Utilize pyranometer, solarimeter, Pyrheliometer and sunshine recorder for measuring Solar radiation.	2	Applying
M4.02	Experiment with digital anemometer & cup anemometer to measure wind speed.	1	Applying
M4.03	Develop a circuit to plot the V-I characteristics of a solar cell.	3	Applying
M4.04	Develop a circuit to plot the V-I characteristics of solar cells in series and in parallel combinations	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Laboratory Manual on Electrical Measurements & Electrical Machines, P.O.Kuttappan, Spades publishers
R1	Electrical Measurements & measuring instruments by A.K.Sawhney DhanpatRai Publications

R2	Electrical Measurements & measuring instruments by R.K.Rajput, S.CHAND Publications
R3	Electrical Measurements & Measuring Instruments by J. B. Gupta, S. K. Kataria & Sons Publications

Online Resources:

Sl.No	Website Link
1	http://vlabs.iitkgp.ernet.in/asnm/#
2	http://www.nitttrchd.ac.in/sitenew1/app_sc/list/Solar%20Cell%20Characteristics.pdf
3	solar cell experiment https://youtu.be/pYIDJ3a8x6E
4	http://www.ee.iitkgp.ac.in/assets/file/es1.pdf
5	https://www.electrical4u.com/electrical-engineering-articles/measurement/
6	https://youtu.be/xQopheOsIpc
7	https://youtu.be/w8Tw2BYSER0

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on an open ended experiment of their choice.

Example 1: Calibrate instruments in the laboratory and prepare error charts.

Example 2: Calibrate Wattmeter/ Energy meter at 0.5 pf lag/lead, 0.866 pf lag/lead by phantom loading.

Example 3: Conduct an experiment to measure the efficiency of solar cell and check the effect of wavelength and angle on efficiency.